

Erwin Quanten

Staff Software Engineer – Python, Node.js, FastAPI, NestJS, AWS

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PROFESSIONAL SUMMARY

Staff Software Engineer with deep experience building **Python 3.8–3.12** and **Node.js 16–20** platforms for cloud infrastructure, telecom, manufacturing, chemicals, and enterprise SaaS industries, delivering secure backend services, event-driven APIs, workflow automation, and production observability across AWS, Kubernetes, PostgreSQL, MongoDB, Redis, RabbitMQ, FastAPI, Django, Express.js, and NestJS. Known for modernizing legacy systems, migrating monolith services to modular microservices, resolving complex production bugs, improving API reliability, and reducing operational waste while leading engineers through practical architecture decisions and release discipline.

SKILLS

- **Languages:** **Python 3.8–3.12**, **Node.js 16–20**, TypeScript 4–5, JavaScript ES6+, SQL, Bash
- **Backend:** **FastAPI 0.95–0.110**, Django 3.2–4.2, Flask 2.x, Express.js 4.x, NestJS 8–10, REST API, GraphQL, WebSocket
- **Databases:** PostgreSQL 12–16, MongoDB 4.4–7.0, MySQL 8, Redis 6–7, Elasticsearch 7–8
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Terraform, GitHub Actions, GitLab CI, Jenkins, Nginx, Linux
- **Messaging & Architecture:** RabbitMQ 3.x, Kafka 2.8–3.x, Celery 5.x, microservices, event-driven systems, distributed tracing
- **Testing & Quality:** Pytest, Jest, Mocha, Supertest, Cypress, unit testing, integration testing, contract testing
- **Monitoring:** CloudWatch, Prometheus, Grafana, ELK, OpenTelemetry, Sentry

EXPERIENCE

Lightedge — Lead Software Engineer

Remote/ Full time

May2022 – Apr 2026

Houston, TX

- Led backend modernization for a cloud infrastructure platform using **Python 3.10–3.12**, **FastAPI**, **Node.js 18–20**, and **NestJS 9–10**, helping engineering teams replace fragile legacy services with scalable API modules.
- Designed secure customer provisioning workflows across **AWS**, Kubernetes, PostgreSQL, and Redis, improving service activation speed by **27%** through cleaner orchestration, retry handling, and async background processing.
- Migrated older Express.js services into structured **NestJS 10** modules with dependency injection, guards, interceptors, and shared validation layers, reducing duplicated backend logic by **23%**.
- Resolved recurring production timeout issues caused by unoptimized PostgreSQL joins, missing indexes, and slow API aggregation, improving high-traffic endpoint response time by **31%**.
- Built **FastAPI** services for internal automation, billing validation, and cloud resource synchronization, giving operations teams faster tools for customer support and infrastructure checks.
- Implemented RabbitMQ-based async processing for provisioning events, usage reports, and account notifications, reducing failed background jobs by **22%** with retry queues and dead-letter handling.
- Improved authentication and authorization using JWT, OAuth2 flows, role-based access control, and API gateway policies across Python and Node.js services.
- Created structured logging, correlation IDs, and OpenTelemetry traces across distributed services, helping engineers diagnose production incidents faster and reduce repeated debugging work.
- Refactored Docker images and CI pipelines for Python and Node services, reducing build time by **19%** through dependency caching, smaller base images, and parallel test execution.
- Partnered with DevOps teams to deploy backend services through GitHub Actions, Terraform, Kubernetes manifests, and AWS managed services with safer rollback procedures.
- Mentored backend engineers on **Python**, **Node.js**, testing strategy, code review discipline, service boundaries, and pragmatic architecture tradeoffs for production systems.
- Implemented automated test coverage with Pytest, Jest, and integration test containers, preventing regression issues in provisioning, billing, and customer account workflows.

Syscom Digital — Staff Software Engineer

Remote/ Full time

Nov 2017 – Apr 2022

Bucharest, Romania

- Built enterprise telecom and digital workflow platforms using **Python 3.7–3.10**, **Django 3.x**, **Node.js 12–16**, **Express.js 4.x**, PostgreSQL, MongoDB, Redis, and RabbitMQ.

- Designed API services for customer onboarding, order tracking, identity workflows, and notification processing, improving internal workflow completion speed by **26%**.
- Migrated legacy Node.js callbacks and scattered middleware into cleaner Express.js service layers with async/await, centralized error handling, and consistent validation rules.
- Resolved memory leak issues in long-running Node.js workers by fixing unclosed database cursors, uncontrolled Redis listeners, and duplicated event subscriptions.
- Introduced **Django REST Framework** services for admin operations and reporting, giving business users faster access to operational data without manual database queries.
- Improved RabbitMQ consumer reliability by adding retry policies, poison-message handling, and idempotent processing, reducing failed message reprocessing by **21%**.
- Optimized PostgreSQL queries, database indexes, and pagination strategy for large customer datasets, improving dashboard load performance by **29%**.
- Implemented secure API authentication with OAuth2, JWT, refresh tokens, scoped permissions, and audit logs across Python and Node.js backend services.
- Created reusable backend libraries for validation, logging, HTTP clients, and error contracts, helping teams build new services faster with consistent behavior.
- Reviewed pull requests, guided architecture decisions, and supported production releases while balancing delivery speed with maintainable backend design.

Tessengerlo Group — Senior Software Engineer

Remote/ Full time

Aug 2015 – Oct 2017

Brussel Metropolitan Area

- Developed manufacturing and chemicals industry applications using **Python 3.5–3.7**, **Django 2.x**, **Node.js 8–12**, Express.js, PostgreSQL, Redis, and Linux-based deployment workflows.
- Implemented APIs for inventory visibility, plant reporting, production approvals, and supplier data exchange, improving operational data accuracy by **24%**.
- Migrated older Python 2 scripts and manual batch jobs into Python 3 services with scheduled execution, structured logging, and safer exception handling.
- Fixed production data mismatch issues caused by timezone conversion errors, stale cache entries, and inconsistent validation between frontend and backend services.
- Built Express.js APIs for internal reporting tools and operational dashboards, giving plant managers faster access to production status and exception reports.
- Improved PostgreSQL schema design, stored queries, and transaction handling for production workflows, reducing duplicate records and manual correction effort by **18%**.
- Implemented Redis caching for repeated lookup data and slow reporting queries, improving dashboard performance by **28%** during peak business hours.
- Supported CI improvements with Jenkins, Docker, automated unit tests, and environment-based configuration for safer backend deployments.
- Collaborated with manufacturing stakeholders, QA analysts, and infrastructure teams to translate complex operational rules into reliable backend workflows.

Tessengerlo Group — Software Engineer

Hybrid/ Full time

Sep 2011 – Jul 2015

Brussel Metropolitan Area

- Built internal business applications using **Python 2.7–3.4**, early Django versions, **Node.js 0.12–6**, Express.js, PostgreSQL, MySQL, and Linux shell automation.
- Developed backend modules for purchasing, logistics, reporting, and user administration, improving internal request processing speed by **22%**.
- Migrated spreadsheet-driven workflows into web-based applications with database validation, approval status tracking, and controlled access for different business teams.
- Resolved recurring deployment issues caused by inconsistent environment variables, missing package versions, and manual server configuration differences.
- Created REST endpoints with Express.js and Python services to integrate internal systems, supplier data, and reporting tools used by operations teams.
- Improved database reliability by adding constraints, safer migrations, backup checks, and validation rules that reduced manual data cleanup by **17%**.
- Implemented background scripts for file imports, scheduled reports, and data synchronization between legacy systems and newer internal applications.
- Worked flexibly across backend development, database troubleshooting, Linux support, release preparation, and user issue investigation when team priorities shifted.

Tessenderlo Group — Software Developer

Hybrid/ Full time

Jun 2009 – Aug 2011

Brussel Metropolitan Area

- Developed early internal web tools using **Python 2.6–2.7**, JavaScript, jQuery, PHP maintenance scripts, MySQL, PostgreSQL, Apache, and Linux servers.
- Created business reporting screens, data-entry workflows, and small automation utilities that helped departments reduce repetitive manual work by **20%**.
- Fixed application bugs related to form validation, broken SQL queries, character encoding issues, and inconsistent browser behavior in older internal systems.
- Supported migration of manual Access and Excel-based processes into centralized web applications with cleaner database storage and user permission controls.
- Worked closely with senior engineers and business users to understand operational problems, test fixes carefully, and deliver stable improvements for daily users.

EDUCATION

KU Leuven — Master's Degree, Computer Science

Sep 2003 – May 2009